

Activity assays were performed to determine the best purification method for a crude, cell-free extract of Protein X. The crude extract was separated into three samples of 800 mg each, and each sample was purified by a different technique. The results are shown in the table below.

Sample	Specific Activity (u/mg protein)	Total Protein Mass (mg)
Crude extract	500	800
Ammonium sulfate precipitation	750	500
Column chromatography	2,000	150
Native PAGE	2,500	5

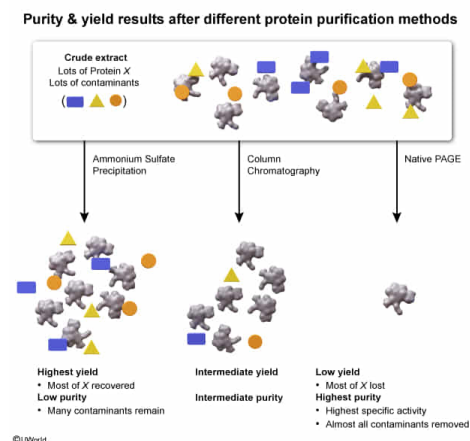
Based on the data, which of the following is (are) true?

- I. Ammonium sulfate precipitation provided the highest total yield of Protein X of the purification techniques.
 - II. Column chromatography provided the highest percent yield of Protein X relative to crude extract.
 - III. Native PAGE resulted in the highest purity of Protein X.
- ☐ A. III only [24%]
☒ B. I and III only [54%]
☐ C. II and III only [15%]
☐ D. I, II, and III [5%]

Correct

54% answered correctly

Explanation:



Extracts from cells are complex mixtures that contain many different proteins, lipids, and carbohydrate molecules. To isolate a specific enzyme, purification methods such as native gel electrophoresis, precipitation, and column chromatography are needed. The effectiveness of each purification method or step is confirmed by performing activity assays that measure the rate of the reaction catalyzed by the enzyme of interest.

In any purification step, some of the desired protein is lost. The **total yield** achieved by a purification step represents how much desired protein (in this case, protein X) is recovered. Yield can be determined by calculating the **total activity** of the purified protein. The **activity** values in an assay have arbitrary units (u) that represent some functional activity of the protein (eg, the catalytic reaction rate for enzymes):

$$\text{Total activity(TA)} = \text{total protein(mg)} \times \text{specific activity}\left(\frac{\text{u}}{\text{mg}}\right)$$

Based on the given table, the TA of each purification technique can be calculated as:

$$\begin{aligned}\text{TA of ammonium sulfate precipitation} &= 500 \text{ mg} \times 750 \frac{\text{u}}{\text{mg}} = 375,000 \text{ u} \\ \text{TA of column chromatography} &= 150 \text{ mg} \times 2000 \frac{\text{u}}{\text{mg}} = 300,000 \text{ u} \\ \text{TA of native PAGE} &= 5 \text{ mg} \times 2500 \frac{\text{u}}{\text{mg}} = 12,500 \text{ u}\end{aligned}$$

Therefore, the ammonium sulfate technique provides the highest total yield of the three methods used **(Number I)**.

However, a high yield of the desired protein does not mean the protein is pure (ie, that undesired proteins were removed). Protein *purity* is represented as **specific activity** (u/mg), which is a **ratio** of the activity of the desired protein to the total mass of protein obtained in a purification step. Consequently, specific activity *increases* during purification as unwanted protein is removed. Based on the table, native PAGE gives the highest specific activity (2,500 u/mg); thus, native PAGE produced the highest purity of Protein X **(Number III)**.

(Number II) To determine the efficiency of a purification technique, the **percent yield** can be calculated as the ratio of purified protein activity to unpurified protein activity. Because all three purification techniques started with the same amount of crude extract, the technique with the highest total yield (ammonium sulfate precipitation, *not* column chromatography) will also have the highest percent yield.

Educational objective:

Specific activity, which reflects protein purity, is the ratio of the activity of the target protein to the total mass of protein (u/mg). The total activity is the product of the total protein mass and the specific activity. Yield is determined by dividing the total activity of the purified protein by the total activity of the unpurified extract.

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